

Control work 11, and work on the mistakes

Systems of two linear equations, by both methods, and the problems behind them.

LESSON 153–154

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 11

EXTENSION REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Solve a system by substitution and by elimination.
- Say from the algebra when there is no solution or infinitely many.
- Turn a two-condition problem into a system and answer both parts.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 432–461
Cambridge	Stage 8 extension
Workbook	Control paper M11

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Solve $\{ y = x - 1 ; x + y = 5 \}$ by substitution	4	L147– 149
2	Solve $\{ x + y = 9 ; x - y = 3 \}$ by elimination	4	L147– 149
3	Solve $\{ 2x + 3y = 12 ; x - y = 1 \}$	5	L147– 149
4	How many solutions has $\{ y = 4x - 1 ; y = 4x + 2 \}$? Explain	3	L145– 149
5	Two numbers add to 26 and differ by 8: find them	4	L150– 152
6	A boat does 22 <i>km/h</i> downstream and 14 <i>km/h</i> upstream: find both speeds	5	L150– 152

WHERE THE MARKS ACTUALLY GO

Q1 one mark for back-substituting rather than stopping at x ; Q2 two for lining the equations up correctly; Q4 two for the words “parallel lines, so no solution”; Q5 and Q6 one each for the “let” sentences and one each for answering both parts.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
substituted into its own equation	$0 = 0$	put it into the other one
only one unknown found	$x = 3$	$(3; 2)$
right-hand sides not subtracted	$2x = 7$	$2x = 7 - 4$
multiplied one side only	$3x - 3y = 1$	$3x - 3y = 3$
“no solution” called an error	the working abandoned	$0 = 3$
unknowns not declared	two equations with no “let”	say what x and y are
only one quantity reported	“the boat does 18”	both the boat and the current

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

IDEA	THE SENTENCE
one linear equation in two unknowns	its solutions fill a straight line
a system	the pairs satisfying both equations at once
graphically	the crossing point of the two lines
substitution	express one unknown, put it into the other equation
elimination	add or subtract to cancel an unknown
$0 = 3$	parallel lines — no solution
$0 = 0$	the same line — infinitely many

LOOKING FORWARD

The last block of the year is combinatorics: counting arrangements and selections. It needs no equations at all — only the two rules of “and” and “or”.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $\{2x + 3y = 12 ; x - y = 1\}$.

- ① From the second, $x = y + 1$.
A coefficient of 1.
- ② $2(y + 1) + 3y = 12$
- ③ $5y = 10$, so $y = 2$.
- ④ $x = 3$. Check: $6 + 6 = 12$ ✓ and $3 - 2 = 1$ ✓

ANSWER

$(3; 2)$

EXAMPLE 2 Model answer, Q4: $\{y = 4x - 1 ; y = 4x + 2\}$.

- ① Setting the right sides equal: $4x - 1 = 4x + 2$.
- ② $0 = 3$
- ③ That is false for every x .
- ④ The gradients are equal and the intercepts differ: parallel lines, no solution.

ANSWER

No solution

EXAMPLE 3 Model answer, Q6: downstream 22, upstream 14.

- 1 Let x be the boat's own speed and y the current's.
- 2 $x + y = 22$ and $x - y = 14$.
- 3 Add: $2x = 36$, so $x = 18$.
- 4 $y = 4$. Both speeds are asked for, so both are given.

ANSWER

Boat 18 km/h, current 4 km/h

03 Interactive model

Return Q5 and Q6 first and count how many scripts reported only one number; the habit of answering the whole question is worth more marks than any technique.

The chapter in twelve questions

$x + y = 5$ alone has:

one solution

no solutions

a line of solutions

two solutions

The solution of a system is:

a number

an ordered pair

a line

an equation

Graphically it is:

the intercept

the crossing point

the origin

the gradient

$\{ y = x - 1 ; x + y = 5 \}$ gives:

$(3; 2)$

$(2; 3)$

$(4; 3)$

$(1; 4)$

Adding $x + y = 9$ and $x - y = 3$ gives:

$2x = 12$

$2y = 6$

$2x = 6$

$x = 12$

So the solution is:

$(6; 3)$

$(3; 6)$

$(12; 3)$

$(6; 6)$

$\{ 2x + 3y = 12 ; x - y = 1 \}$ gives:

$(2; 3)$

$(3; 2)$

$(1; 2)$

$(4; 3)$

Equal gradients and different intercepts give:

one solution

no solution

infinitely many

two solutions

Equal gradients and equal intercepts give:

one solution

no solution

infinitely many

none of these

Two numbers adding to 26 and differing by 8 are:

17 and 9

16 and 10

18 and 8

15 and 11

Downstream 22 and upstream 14 give a current of:

4

8

18

36

When a problem names two quantities you should report:

one of them

both of them

the equations

x only

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
System	Sistema	Система
Substitution	O'rniga qo'yish	Подстановка
Elimination	Qo'shish usuli	Способ сложения
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
No solution	Yechimga ega emas	Нет решений
Infinitely many	Cheksiz ko'p	Бесконечно много
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 finishes with:

Q2 uses:

Q3 is quickest by:

Q4 answer is:

one solution

no solution

infinitely many

$(0; 3)$

Q5 needs:

one number

two numbers

an equation

a graph

Q6 asks for:

the boat only

the current only

both speeds

the distance

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\{ y = x - 1 ; x + y = 5 \}$

ANSWER (3; 2)

2. $\{ x + y = 9 ; x - y = 3 \}$

ANSWER (6; 3)

3. $\{ 2x + 3y = 12 ; x - y = 1 \}$

ANSWER (3; 2)

4. $\{ y = 4x - 1 ; y = 4x + 2 \}$

ANSWER No solution

5. Two numbers adding to 26 and differing by 8

ANSWER 17 and 9

6. Downstream 22, upstream 14: the boat

ANSWER 18

7. And the current

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. $\{ 3x + y = 11 ; x + y = 5 \}$

ANSWER (3; 2)

2. $\{ y = 2x + 1 ; y = 3x - 2 \}$

ANSWER (3; 7)

3. $\{ 2x + y = 8 ; 3x - y = 7 \}$

ANSWER (3; 2)

4. $\{ y = 4x - 1 ; 2y = 8x - 2 \}$

ANSWER Infinitely many

5. A rectangle of perimeter 48, length 6 more than width

ANSWER 9 by 15

6. Complementary angles differing by 20°

ANSWER 55° and 35°

7. $\{ x + 2y = 7 ; x = 3 \}$

ANSWER (3; 2)

Hard · stretch and reason

7 PROBLEMS

1. $\{ 2x + 5y = 16 ; 3x - 2y = 5 \}$

ANSWER (3; 2)

2. $\{ 4x + 3y = 18 ; 2x - y = 4 \}$

ANSWER (3; 2)

3. Digits add to 12, reversal 18 larger

ANSWER 57

4. 4 notebooks and 3 pens cost 26 000; 2 notebooks and 1 pen cost 12 000

ANSWER Notebook 5000, pen 2000

5. For which k has $\{ y = kx + 1 ; y = 5x - 2 \}$ no solution?

ANSWER $k = 5$

6. Why is " $x = 3$ " an incomplete answer to a system?

ANSWER The solution is a pair

7. $\{ x - y = 0 ; x + y = 14 \}$

ANSWER (7; 7)

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Solve $\{ y = 2x - 3 ; x + y = 6 \}$ by substitution.
3. Solve $\{ 3x + 2y = 16 ; 3x - 2y = 8 \}$ by elimination.
4. Say how many solutions $\{ y = 5x + 2 ; 2y = 10x + 4 \}$ has and why.
5. Two numbers add to 34 and differ by 10. Find them.